Q1. What is 127.0.0.1 ?

- It is a special IP address
- It means this computer itself
- This is called the loopback address
- Suppose your computer has this IP to your another computer
- So the request does not go to another computer on the network
- It comes back to the same computer
- If you open http://127.0.0.1:8080 , this means connect to port 8080 on this same computer

Q2. Is 127.0.0.1 a special IP address ?

- This belongs to the IPV4 loopback address range : 127.0.0.0/0
- This commonly used address has 127.0.0.1

Like :

```
http://127.0.0.1:3000
http://127.0.0.1:8080
http://127.0.0.1:5000
```

- These usually mean run or connect to the application on my own computer

Q3. What is localhost?

- This is a name that normally refers to your own computer
- So we use localhost with 127.0.0.1 and also same as http://localhost:8080 which similar to http://127.0.0.1:8080
- These usually reach the same computer
- So localhost is the name meaning this computer
- So 127.0.0.1 is the IP address meaning this computer
- So loopback means that network communication back to the same computer

Q4. What is the relationship between localhost and 127.0.0.1 ?

- This is like a person name and phone number
- So localhost is the name and IP is address such as 127.0.0.1
- When computer sees localhost , it goes to loopback address such as 127.0.0.1
- Therefore localhost:8080 means 127.0.0.1:8080

Q5. What is a port ?

- So an IP address identifies which computer
- So an port identifies which application or service on that computer
- Such as 127.0.0.1:8080 means which computer or this computer using 8080 which application or service where application listening on port 8080
- So react frontend uses : port 3000 , and spring boot uses port 8080 and FastAPI uses port 8000
- So http://localhost:3000 could be react application and localhost:8080 can be your spring boot backend

Q6. What is HTTP ?

- Hypertext Transfer Protocol
- Protocol used for communication between a client and a server
- So browser uses http request to access web server and based on the HTTP response we can see what we have requested using browser
- So http://example.com is the browser communicates with the server using HTTP

Q7. What is HTTPS ?

- HTTP means HTTP Secure
- An HTTP communication protected using TLS (Transport Layer Security)
- So using http with browser is a readable traffic to server
- Whereas using https with browser is encrypted while reading or using the server
- So HTTPS provides important security properties like Encryption , Authentication of the server and protection

Q8. Comparison between HTTP vs HTTPS

HTTP HTTPS

HTTP	HTTP + TLS security
Port 80	Port 443
Traffic not encrypted by HTTP	Traffic is encrypted with TLS
Someone on the network may be able to read / modify traffic	TLS protects the traffic from network to avoid eavesdropping or tampering
No server certificate authentication from HTTP	Server identity is authenticated using certificates
http://	https://

Q9. Why is HTTP considered not secure ?

- Suppose you log into a website using HTTP like : http://example.com/login
- So your browser sends information across the network
- So HTTP does not encrypt the communication
- So we use username and password to access internet using router to access website using HTTP
- So an attacker who can observer the network traffic may see the contents
- So that why HTTP is considered insecure for sensitive communication

Q10. What happens with inside HTTPS ?

- With HTTPS , while using router to access website where the traffic is encrypted both sides between us and the website to access with router
- So TLS protects this
- Instead of someone seeing username and password which is openly readable by using HTTPS the network traffic appears as encrypted data
- So HTTPS prevents ordinary network observers from reading the protected HTTP contents

Q11. Why is HTTPS Secure ?

- So HTTPS uses TLS
- TLS establishes a secure connection between your browser and the website
- So the browser connects to the website and website provides a digital certificate helps the browser verify that it is communicating with the real and safe website
- So the browser and server establishes shared cryptographic keys where the communication comes encrypted with TLS while using browser to access server
- So HTTP messages are carried inside the protected TLS connection

Q12. What does the HTTPS certificate do ?

- Suppose we visit a website and our browser needs to know that server is seeing real website
- So the sites TLS certificate helps establish the server identity
- So a trusted Certificate Authority known as CA has issued and signs the certificates
- So website uses TLS certificate with trusted CA and browser trusts the CA where the browser can validate the certificate

Q13. Why does HTTPS protect against someone changing the data ?

- So TLS provides protection
- Imagine you request transfer like 100 rupees
- So an attacker shouldn't able to change the message to transit to transfer 10,000 rupees
- So without the TLS protection the protected data can be modified
- So using HTTPS with encryption where it protects confidentially while authentication which helps verify the server to protect tampering of data

Q14. Why does HTTP show Not Secure ?

- Suppose you type http://example.com
- Your browser knows this connection isn't protected by HTTPS and TLS
- Therefore modern browsers may display as not secure or another warning indicator
- This means the connection is not protected by HTTPS

Q15. Why can you still click continue to site ? (HTTP)

- So HTTP is allowed by browsers
- If we use http://example.com and the browser can say this connection is not secure
- But it can allow you visit the site , because HTTP itself is a legitimate protocol and can be used for development testing , redirects and non-sensitive resources
- So this warning is telling you about the security of the connection and not blocking access to the website

Q16. Can HTTPS also produce a warning ?

- So HTTPS does not mean everything is okay
- So https://example.com could have an invalid or expired or mismatched certificate
- So the browser can show something like your connection is not private
- Because the browser cannot properly validate the sites TLS certificate
- So using HTTP with no TLS protection is unsafe
- So using HTTPS + valid TLS protected connection is safe
- So HTTPS + certificate having TLS problem using browser can be warning situation

Q17. Why do we use HTTP on localhost ?

- You might use and run : http://localhost:3000 instead of https://localhost:3000
- Because you are the developing the application on your own computer
- So you are communicating locally and every development setting up TLS certificate for very local application is difficult and not necessary
- Therefore , developers commonly use :

http://localhost:3000 http://localhost:8080 http://localhost:8000

Q18. Is HTTP localhost secure ?

- So localhost is special because traffic addressed to the loopback interface which stays on our local machine under normal network behavior
- So browser can use HTTP with 127.0.0.1 to access local application
- So this isn't applicable using browser using HTTP to access internet using remote server
- So HTTP on localhost is common for developers or development

Q19. Important misconception ?

- So localhost is actually mean secure and exactly tells where the server is like this computer
- So HTTP and HTTPS tells how the communication is protected
- So HTTP uses with No TLS and HTTPS is used with TLS protection

Q20. Model ?

- So we use the IP address such as 127.0.0.1 as where which Is this computer
- So we use the port such as 8080 is which application
- So using HTTP such as http:// is how we communicate without TLS encryption
- So using HTTPS such as https:// is how we communicate using protected TLS
- So Finally , we use :

http://localhost:8080/api/tasks

http	Protocol
localhost	This computer
8080	Application port number
/api/tasks	API endpoint

- So therefore, We use HTTP to communicate with the application running on port 8080 on this computer and request using /api/tasks

So when we use :

https://example.com/api/tasks

https	HTTP protected by TLS
example.com	Remote website or server
/api/tasks	API endpoint

Q21. HTTP vs HTTP Security ?

HTTP Browser ——————————→ Server readable traffic ❌ No TLS encryption ❌ No TLS server authentication ❌ No TLS integrity protection

HTTPS

Browser ═══════════════════════════→ Server
TLS protected

- ✅ Encryption
- ✅ Server authentication
- ✅ Integrity protection

127.0.0.1 = This computer

localhost = Name for this computer

IP address = Which machine/interface?

Port = Which application?

HTTP vs HTTPS = How securely the web communication is carried

- So therefore , localhost and HTTPS are not the same thing , localhost tells you where the application is and HTTPS tells you how the communication is protected

So ,

`http://localhost:8080`

- Means : Talk to my computer applications on port 8080 using HTTP

While :

`https://example.com`

- Means : Talk to the remote website using HTTP protected by TLS (HTTP + S) (HTTPS)